



EFF

BadgerInfo++

BadgerInfo++: A Layered-Explanation Extension for Web-Tracking Transparency

Dealing with Targeted Advertisements

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Master’s Project



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Abstract

This study investigates how different explanation formats such as textual, visual, and layered presentations affect users' understanding of online tracking and targeted advertising. A modified version of the **Privacy Badger** extension, called **BadgerInfo++**, was developed to provide contextual explanations about covert tracking through multiple feedback modes. A qualitative user study involving **38** participants was conducted using semi-structured interviews. Participants interacted with various explanation styles and reflected on their experience with the tool. Results indicate that users valued having control over the depth of information. Non-technical users preferred simple textual summaries, while technically inclined users favored visualizations and detailed breakdowns. Layered explanations were particularly effective in accommodating these diverse preferences. Findings suggest that explanation style significantly influences perceived transparency, user engagement, and confidence in managing personal data. Privacy tools that incorporate user-tailored educational features, in addition to blocking functionality, may enhance user awareness and trust.

Keywords: privacy, web tracking, browser extension, layered explanations, targeted advertising, transparency, user study, HCI

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Chapter 1

Introduction

In today's digital ecosystem, user data has become a valuable commodity. Behind the scenes of everyday web browsing, a complex and often invisible network of online trackers collects personal information - ranging from browsing habits to device identifiers - without users' explicit knowledge or consent. Despite growing public discourse around digital privacy, many users remain unaware of how, when, and why their data is being collected [16]. This lack of awareness presents a critical challenge to meaningful user agency and informed consent in online environments.

Privacy-enhancing technologies (PETs), such as tracker-blocking browser extensions, offer users tools to mitigate unwanted surveillance. However, these tools often focus on functionality - blocking scripts, disabling cookies, or minimizing data sharing - while neglecting the equally important goal of user education. When users do not understand how these tools work or what threats they are protecting against, the transparency and trustworthiness of the privacy tool can be compromised. Simply put, a tool that works invisibly may protect data, but it does little to empower users.

This project is situated at the intersection of privacy technology and user experience. The team set out to explore how the presentation of information, that is, how privacy tools explain themselves and the threats they counteract, can shape users' understanding and engagement. Rather than focusing solely on what is blocked, this project's interest lies in how different explanation styles influence awareness, perception, and behavioral intention.

This motivates the central research question:

How do different levels or types of explanations about covert tracking affect users' understanding and preferences?

1.1 Background

Concerns about privacy have long existed in the digital space. While people's understanding of targeted advertising is often shaped by their personal experiences, it remains limited

in scope [13].

Our research seeks to address these issues by combining technical transparency with engaging, narrative-driven feedback. To investigate this, **Privacy Badger**, a well-known open-source browser extension developed by the Electronic Frontier Foundation (EFF), was selected. Privacy Badger blocks third-party trackers automatically and is widely regarded for its effectiveness. However, like many PETs, it offers limited in-depth feedback to users beyond showing which domains have been blocked. While this simplicity is beneficial for usability, it may hinder deeper understanding of covert tracking behaviors, especially for less technically inclined users.

Recognizing this, rather than simply blocking trackers, the extension aims to:

- Detect sensitive data and flag potentially intrusive tracking practices.
- Visually and textually explain how user data is collected and transmitted.
- Empower users by improving their understanding of targeted advertising, enabling them to make more informed decisions about their privacy.

By building on Privacy Badger’s core functionality, the project aims to enhance both the technical robustness and the educational impact of privacy protection tools.

1.2 Problem Statement

Despite the availability of various anti-tracking tools and privacy extensions, many users still lack a clear understanding of how targeted advertising work and, by extension, how their personal data is being collected and used [16]. This gap in understanding undermines the effectiveness of privacy tools by reducing user trust, limiting perceived value, and weakening informed digital decision-making. Specifically, the issues include:

- **Limited User Awareness:** Users are often unaware of the extent and mechanisms of data collection behind targeted ads.
- **Insufficient Transparency:** Existing tools sometimes offer only basic data-blocking functionalities without explaining the processes, leaving users confused about what is being tracked and why.
- **Engagement Deficit:** Without intuitive and engaging explanations, users are less likely to adjust their privacy settings or take an active role in managing their digital footprint.

These challenges necessitate a tool that not only detects and reports on data collection but also translates these technical details into accessible, narrative-driven explanations that empower users and promote greater control over their online privacy.

1.3 Objectives

The study and extension development are guided by the following objectives:

1. Extension Development and Integration:

- **Build on Privacy Badger:** Extend its functionality by adding features that visualize sensitive data flows and provide narrative explanations.
- **Integrate Tracking and Visualization:** Incorporate real-time data flow monitoring and interactive visualizations that clearly demonstrate how user data is tracked.
- **User-Friendly Feedback:** Design the extension interface so that even non-expert users can easily understand technical terminology and data tracking practices.

2. User-Centered Evaluation:

- **Conduct a User Study:** Use surveys, interviews, and controlled usability testing to assess how users perceive and engage with the extension.
- **Assess Educational Impact:** Compare different explanation formats (text-only, visual, and combined) to identify the most effective method for enhancing user understanding.

3. Educational and Design Impact:

- **Inform Best Practices:** Generate insights into how privacy tools can be refined to balance technical protection with educational value.
- **Empower Users:** Ultimately, provide users with actionable information to better manage their online privacy.

Chapter 2

Related Work

2.1 Targeted Ads and Tracking: What are trackers, how do they work, why is this a problem?

Online Behavioral Advertising (OBA) is the practice of tailoring advertisements to individual users based on their browsing behavior and personal data. It relies on tracking technologies such as cookies, web beacons, browser fingerprinting, and more recently, CNAME cloaking.

Trackers can include:

- **Cookies:** Store browsing history and preferences.
- **Tracking pixels/beacons:** Invisible elements that report user interactions.
- **Browser fingerprinting:** A technique that builds a unique device ID based on hardware/software configurations.
- **CNAME cloaking:** Disguises third-party domains as first-party to evade detection.

These methods allow advertisers to build detailed user profiles and serve personalized ads across websites.

Tracking, although intended to increase ad relevance, raises significant privacy concerns. Users often don't know what data is collected, who collects it, or how it's used. Research has shown that users perceive tracking as invasive, particularly when it seems to involve sensitive or personal data [18]. The growing awareness of surveillance capitalism has also contributed to public unease, especially with the realization that even seemingly benign interactions (e.g., reading an article) can contribute to detailed profiling [1].

In particular, large-scale audits have demonstrated that many browser extensions or scripts collect sensitive page content without explicit consent, sometimes in alarming quantities [14]. Such findings support the urgent need for transparency and user education alongside technical protections.

2.2 Existing Tools

A number of browser extensions exist to help users limit or block third-party trackers. Each varies in terms of functionality, user interface design, and educational value:

Tool	Tracker Blocking	Sensitive-Data-Awareness	User Interface	Educational Features	Open Source
Privacy Badger	Learns to block based on behavior (not lists)	No	Moderate (popup-based)	Minimal (color indicators only)	Yes
Ghostery	Uses tracker lists + categorization	No	High (modern UI)	Moderate (explanatory labels)	Partially
uBlock Origin	Highly effective; list-based blocking	No	Low (technical UI)	None	Yes
Decentraleyes	Blocks CDN-hosted third-party scripts	No	Very Low	None	Yes

Table 2.1: Comparison of Privacy Tools

- **Privacy Badger** (by the Electronic Frontier Foundation) blocks trackers based on behavior rather than static blocklists. It also sends Global Privacy Control and Do Not Track signals. Privacy Badger attempts to strike a balance between blocking invasive trackers and preserving page functionality. However, its interface lacks detailed explanations about what's being blocked or why it matters [12].
 - **Privacy Badger** is most aligned with behavioral learning but doesn't offer insight into what data was accessed or why.
- **Ghostery** provides users with a categorized list of trackers on a given site (e.g., analytics, social, advertising) and offers optional blocking. While more user-friendly in terms of interface, it doesn't visualize how data flows or which personal details may be leaking.
 - **Ghostery** helps with general awareness.
- **uBlock Origin** is a powerful content blocker that applies multiple filter lists to block ads, trackers, and malware domains. Though highly effective, it's primarily geared toward advanced users and lacks educational components or explanatory features [9].
- **Decentraleyes** is more specialized: it prevents sites from loading resources from third-party CDNs (like jQuery from Google), minimizing tracking by large third-party providers. It doesn't deal with sensitive data visibility or provide user-facing explanations.

While these tools block or hide trackers, they rarely explain tracking in ways accessible to average users or make visible the data journey that leads to targeted ads. This is where the project's approach comes in, combining data detection with layered, individual explanations with the aim to innovate.

2.3 Literature Review: User Perception and Privacy Studies

This section reviews existing research on user perceptions of tracking, targeted advertising, and privacy tools. Where relevant, observed patterns from the current study are briefly connected to prior work to situate the design within the broader literature.

2.3.1 User Understanding and the Privacy Paradox

Despite increasing public concern about online privacy, studies consistently show that many users lack a clear understanding of how web tracking works. Pew Research Center [2] found that approximately 79% of Americans felt they had little or no control over the data companies collect about them, and 59% admitted they did not understand how their information was used. This aligns with the common misconception that trackers are merely cookies or visible elements, when in reality, they include hidden scripts, browser fingerprinting techniques, and third-party beacons. This limited comprehension creates a gap between the existence of anti-tracking tools and their effective use. Users may install blockers but remain unaware of what they're actually protecting themselves from.

This lack of understanding is compounded by what is often referred to as the **privacy paradox**. First described in behavioral economics and HCI literature, the privacy paradox refers to the phenomenon where users express concern about their digital privacy but fail to adopt protective behaviors in practice (Acquisti et al., 2015). For instance, users often accept terms and conditions without reading them, reuse passwords, or grant extensive app permissions without scrutiny, despite professing a desire to protect their data. Several factors explain this discrepancy, including habituation, convenience, and the belief that opting out makes little difference [3].

In this context, the privacy paradox reinforces the need for educational tools like **BadgerInfo++** that not only block trackers but also help users understand why they should care. Rather than assuming users will make informed decisions once given the option, tools must bridge the comprehension gap with intuitive explanations. Observations from the user study echo this sentiment: many participants initially underestimated the scale or complexity of tracking but reported increased awareness after using the extension. This suggests that improved literacy, facilitated by layered, accessible explanations, could help realign users' stated privacy values with their actual behaviors.

2.3.2 The Impact of Transparency Interventions

Transparency interventions are features or mechanisms that inform users about how and why their data is being used. Prior research has shown that such mechanisms can significantly influence user engagement and perception in privacy-sensitive contexts. In the domain of social media and online disclosure, even lightweight cues have the potential to prompt more thoughtful user behavior. For example, Wang et al. [19] conducted a field study on Facebook that tested two types of nudges: one displaying profile pictures of people who would see a post, and another introducing a short delay before publishing. These nudges encouraged users to reconsider their posts, with some choosing to edit, delay, or

cancel them. The findings suggest that subtle interface elements can lead users to reflect more actively on privacy decisions.

A related study by Strömel et al. [18] focused on fitness tracking and explored how narrative feedback generated by large language models could enhance user understanding. Participants who received personalized summaries of their fitness data reported increased reflection and engagement compared to those who viewed only raw metrics. Although situated in the context of health data, the study highlights the potential of contextual, user-tailored explanations to support digital literacy across domains. This aligns with prior work in usable privacy emphasizing that tools must be not only available but also understandable and actionable [11].

The design of **BadgerInfo++** incorporates these principles by embedding layered explanations within the browser extension interface. Rather than presenting all technical information upfront, the extension allows users to choose the level of detail they wish to explore. During the user study, several participants noted that even minimal explanations helped them reflect on the presence and purpose of web trackers. These observations support broader claims in human-computer interaction research that transparency is most effective when it is interactive, incremental, and context-sensitive [7].

Through a combination of textual summaries, visual aids, and optional in-depth explanations, *BadgerInfo++* demonstrates how transparency interventions can move beyond data protection alone. By promoting user understanding and reflection, such tools contribute to the growing field of privacy-enhancing technologies that aim to educate, inform, and empower.

2.3.3 Reducing Discomfort Through Explanations

One of the key emotional reactions users report in response to targeted advertising is discomfort, often described as the ad being "creepy" or "too personal." This reaction stems from the perceived invasion of privacy, particularly when users are unsure how or why they were targeted in the first place. However, studies have shown that these negative perceptions can be reduced through transparent and contextual explanations that clarify how user data is used in ad delivery.

Eslami et al. [5], for example, found that users were more accepting and reflective when algorithmic decisions were accompanied by explanations. In their study, participants who were informed about why specific items appeared in their Facebook news feed reported greater understanding and less frustration compared to those who received no explanation. These findings underscore the role of transparency in shaping emotional responses to automated systems.

During the evaluation of **BadgerInfo++**, similar patterns were observed. Participants who were initially startled by the number and nature of web trackers became more receptive once they encountered brief, plain-language descriptions explaining each tracker's purpose. Some expressed curiosity and chose to explore the additional details provided in the "Learn More" section. These observations support the notion that discomfort does

not necessarily stem from tracking itself, but from a lack of clarity and control. By bridging this gap, layered explanations can help reduce emotional discomfort and foster more thoughtful interactions with privacy tools.

2.3.4 Narratives and Visualizations for Comprehension

Visual and narrative-based explanations have gained increasing attention in privacy research as effective methods for making complex data practices more understandable to everyday users. While traditional interfaces often present data in raw or technical formats, these alternatives aim to humanize and contextualize information in ways that promote reflection and learning. In particular, narrative feedback can help users form a mental model of how their data is being tracked and used, while visualizations offer accessible overviews of invisible processes.

Strömel et al. [18] explored this concept in the domain of fitness tracking by combining large language models with personal informatics tools. Their system generated narrative-style feedback that interpreted raw step count data in plain language. Participants who received these reflective summaries reported increased engagement with the tool, greater emotional connection to their data, and improved understanding of their behavior over time. Though focused on physical health, the findings underscore the broader potential of narrative explanations to support comprehension in other domains, including privacy.

Similarly, Hull et al. [10] found that data visualizations significantly enhanced users' understanding of online tracking. When participants were shown interactive maps revealing which third-party trackers were following them across websites, their awareness and concern increased. Importantly, users reported that the ability to see relationships between companies and the extent of data sharing made the tracking ecosystem more tangible and less abstract.

The design of **BadgerInfo++** was informed by this literature. The extension includes both brief textual descriptions of individual trackers and an internal page with structured explanations and interactive visualizations. These elements can be explored in layers, allowing users to control how much detail they wish to engage with. Some participants indicated that the visualizations helped them “see what’s really going on,” while others preferred the simplicity of textual summaries. This highlights the value of a multimodal approach in addressing diverse user preferences and literacy levels.

Ultimately, narratives and visualizations serve as cognitive tools that promote understanding, engagement, and decision-making. In the context of online privacy, where tracking mechanisms are often opaque and abstract, such tools play a crucial role in connecting technical systems with user agency.

2.3.5 Tailored Interfaces and Trust

Personalization is not only a tactic used by advertisers but also an effective strategy in the design of privacy-enhancing technologies. Tailoring interfaces to suit different levels of user expertise and interest has been shown to foster greater trust, satisfaction, and

sustained use. This is particularly important in the domain of data tracking and privacy, where users vary widely in their technical understanding and sensitivity to data collection. Designing a one-size-fits-all interface risks alienating both novice and expert users, either by overwhelming the former or under-informing the latter.

Demuth et al.[4] introduced the concept of “support personas” to illustrate how privacy tools can adapt to different user profiles. Their study found that novice users preferred simplified summaries and guiding text, while experienced users valued detailed technical breakdowns and fine-grained control. When users were given the option to choose the type and depth of explanation they received, both groups showed increased trust in the tool and were more likely to explore privacy settings. This underscores a central point in human-computer interaction: user trust grows when systems align with users’ expectations and cognitive styles.

Additional studies reinforce the benefits of tailoring. Wang et al. [19] investigated the impact of interface complexity on privacy decision-making and found that users who were matched with an appropriate complexity level were more accurate in their decisions and felt more in control. Conversely, mismatches between interface style and user type led to confusion, frustration, or even disengagement. These findings point to the importance of adjustable or adaptive features in privacy interfaces.

The design of **BadgerInfo++** incorporates this principle through a system of layered explanations. Users begin at a base level of detail but can choose to incrementally reveal more information, either through additional text, interactive visualizations, or a “Learn More” internal page. This design respects users’ agency and allows them to explore at their own pace. During testing, some users appreciated being able to access more in-depth content when curious, while others were content with the simpler tracker summaries. This confirmed that offering adjustable levels of detail can meet a wider range of needs and expectations.

By tailoring the interface to user capabilities and interests, the goal is not only to inform but also to build a sense of trust and confidence. When users feel that a tool “meets them where they are,” they are more likely to engage with it meaningfully and integrate it into their privacy practices.

Chapter 3

System Design

3.1 Introduction

The main functional component of this project is a Chrome¹ extension which the user can install themselves and use while browsing the web. At any point the user can choose to interact with the extension and learn more about the trackers they have come into contact with, and trackers in general. The extension itself is a modification of the pre-existing extension "Privacy Badger" made by the Electronic Frontier Foundation. The project team removed certain features of the extension, while adding in new ones, and modifying some existing systems. The details of which will be described in the following sections. One important note to mention before continuing is that certain aspects of this extension were created with the aid of a **Large Language Model** (LLM).

3.1.1 EFF's Privacy Badger Extension

As described by the EFF on their website: "Privacy Badger is a browser extension that automatically learns to block invisible trackers"[6]. The main purpose of the base extension is to provide the user a basic "popup" window of information on all the current trackers (domains) on the webpage, as well as granular control over if they should be allowed to function, block their use of cookies, or to be forbidden to run outright. The extension also allows the user to access a Frequently Asked Questions (FAQ) page that answers important questions about trackers and the extension. The extent of the extension's popup window can be seen in the image below, where it is being ran in the context of an **overleaf** paper.

¹While the original PrivacyBadger extension is compatible with the Chrome, Firefox, Edge, and Opera web browsers, the project's extension was made exclusively with Chrome in mind, so may not deliver full functionality on other browsers.

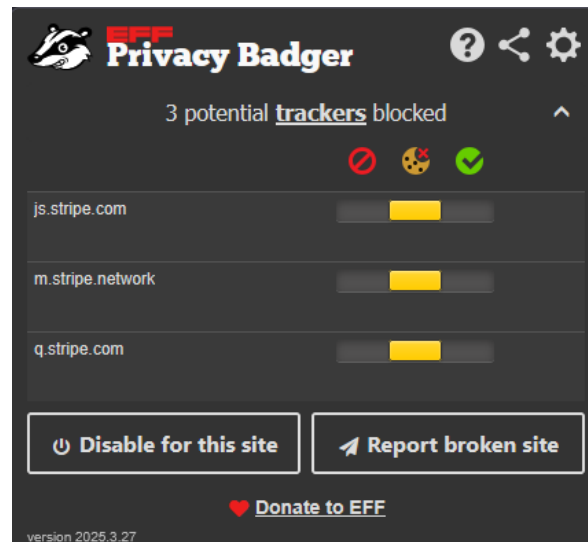


Figure 3.1: The popup window of the unmodified Privacy Badger extension

Privacy Badger is an open-source extension, meaning that its entire codebase (hosted on GitHub) is available online for anyone to view, contribute to, or modify (given any external use abides by the copyright license the owner has applied to it). This was one of the deciding factors for picking this extension as the subject of modification for this paper's research. Although the documentation for the extension was nearly non-existent, free and easy access to the entire (up to date) codebase allowed for smooth modification during the construction process. The modifications of which will be described in the next section.

3.1.2 Overview of Changes

The overall aim in modifying the extension was to provide the user more information about the individual trackers, more detail about what trackers are in general, different methods of visualizing their trackers, and a way to control how much detail they see about said trackers, while also retaining the core features of the base extension. This modified extension is hereafter referred to as "**BadgerInfo++**" or **BadgerInfo** for short.

3.2 BadgerInfo++

3.2.1 Removal of Base Features

An important detail to mention before continuing, is the removal of certain aspects of the pre-existing extension. Though broadly considered as "bloat" by the project team, the main motivation for the removal of some elements of the PrivacyBadger extension was to prevent instances where features and explanations provided by the project team clash with elements (and more importantly explanations) already present on the extension, ones that

may disrupt the actions made by the users of the extension. As an example, the Privacy Badger extension has two instances that link to their FAQ page. The team felt that this would detract from the information page implemented for the extension, and not respect the added control over levels of information the user now has access to.

Overall, the following items were removed from or simplified in the new extension:

- Report broken site button
- Lower left extension "version" text
- Share button
- Help/FAQ button
- Hyperlink on the "trackers" text to FAQ page
- Popup explaining how to log in to Google websites
- First time install popup
- First time install landing page

With these items removed, space can be made for the added features to shine.

3.2.2 Welcome Page

A base functionality of the PrivacyBadger extension is a "Welcome Page" the user automatically sees after installing² the extension. For the purposes of this project, this page was modified and simplified to reflect the information required. Seen below is an image of the welcome page the user will automatically see a few seconds after installation.

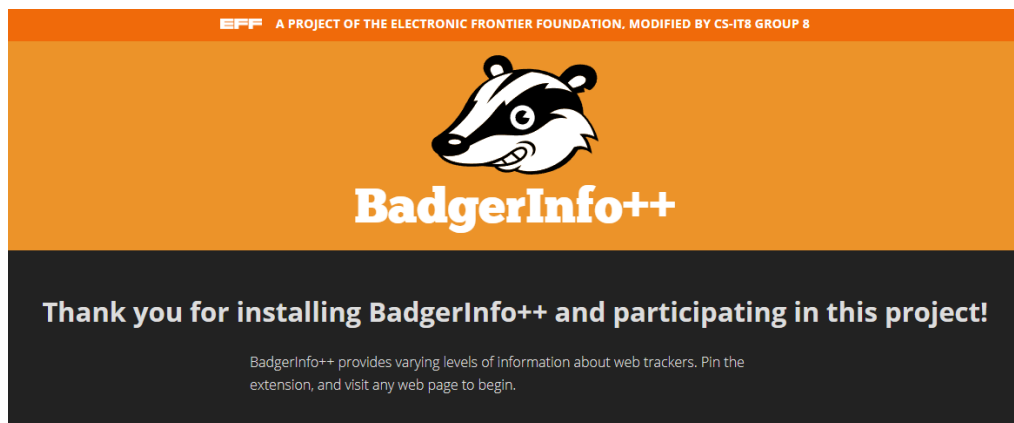


Figure 3.2: The welcome page the user sees after installing the extension.

²Due to the small scale and short timeline of the project, users had to manually install the extension via a zip file, as opposed to a two click installation process that most extensions found on the Chrome web store have.

3.2.3 Notification Indicator

One of the first features the user will come in to contact with is the green "notifier" popup. This appears upon the extension detecting a new tracker on a specific web page. The notification stays on the screen for 5 seconds before disappearing. The notification indicates how many total trackers are on the page, and direct the user to open the extension popup in the top right of the screen. Below is an example of what the popup looks like.

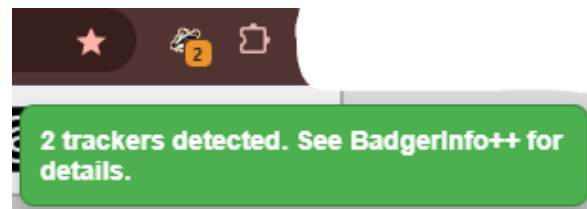


Figure 3.3: The notification that appears in the top-right of the screen upon a tracker being detected.

The notifier works by utilizing one of Privacy Badger's built-in functions *updateBadger* inside the *background.js* file, so that every time the (pre-existing) number on the extension "badge" updates (this number also indicates the number of trackers) then the notifier will appear as well. This does result in instances where the notifier appears multiple times in a row as new trackers are detected, but the team is happy with this unintended side effect as it presents more chances to attract the user's attention to the popup itself.

3.2.4 Levels of Detail

The key mechanism at the heart of the BadgerInfo++ extension is the varying levels of detail the user has access to. Inside the popup, and the later mentioned "Internal Page" the user has access to two arrow buttons that allow them to increase or decrease the amount of information they have presented to them. In total there are technically 6 different levels of detail. Three on the popup, and three on the internal page. By default on installation, the extension starts at the lowest level of detail. This is to aid in one of the main research aspects of this project, to identify what level of information, and of what type, the user likes most. In the following sections the information which can be revealed to the user by modifying the level of detail will be described. The detail adjustment controls can be seen in the images below. As observed in the first image, if the user hovers over a button, a tooltip appears informing them what the button does.



Figure 3.4: The detail control buttons on the popup, and internal page.

The choice to have multiple levels or "variants" of information display was directly inspired by the paper "*Narrating Fitness: Leveraging Large Language Models for Reflective Fitness Tracker Data Interpretation*". In this paper they split their representations into three

types; text only, chart only, and a combination of text and chart [17]. It was believed that by utilizing their method of information separation the team could extract key insights from the users of the extension. At a previous state of the project these different "variants" would be completely separate, and randomly assigned to the user upon install, but it was decided that allowing the user to cleanly switch between them would allow for a more fruitful study.

3.2.5 Popup Window

The "popup" window is one of the main components of this extension. Upon clicking the extension icon in the top-right of the screen this popup window will open. It retains the same layout and some features from the original unmodified *PrivacyBadger* extension, but has had some alterations for this project. Seen below is the popup window as it appears on a fresh installing, and on the website <https://online2pdf.com/docx2pdf>.

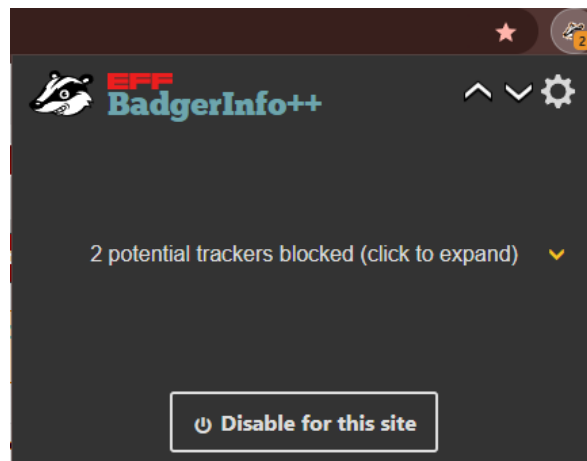


Figure 3.5: The popup window upon a fresh install in use on a website.

If one were to compare this image to the base *PrivacyBadger* popup, it is clear the components that were removed or reduced. It is also visible the surface level modifications.

Upon a default install, the popup begins by displaying the lowest level of information. An example of this can be seen in the image below.

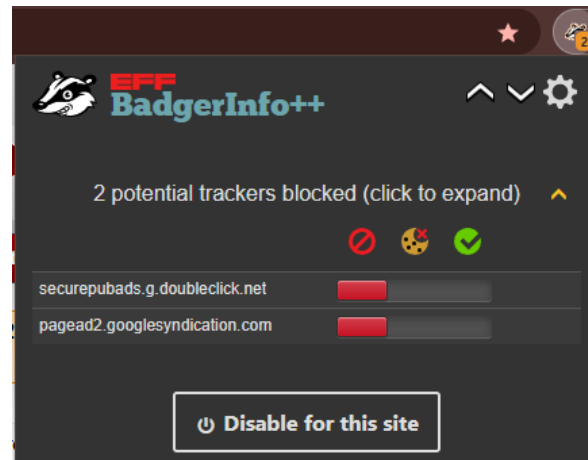


Figure 3.6: The popup window upon a fresh install in use on a website (drop down expanded).

The popup displays the tracker URLs found on the current page, and provides the user with the default PrivacyBadger controls, but beyond this, the information is very limited. The idea behind this "variant" is to be a control case, providing on a minimal amount of details, and to not overwhelm the user immediately. As the user increases the level of detail, more elements will become visible, so they have are eased in to what the extension has to offer.

At the next level of detail, the user gets a single sentence explanation of each tracker URL the popup has found. The goal was to provide the user a quick snippet of information, without overwhelming them, and filling the entire popup window. It is important to note that, whenever a tracker is detected, its URL is saved to list inside *background.js* which will be used later on the **internal page** for the visualization. Seen below is an example of the popup with the individual tracker explanations.

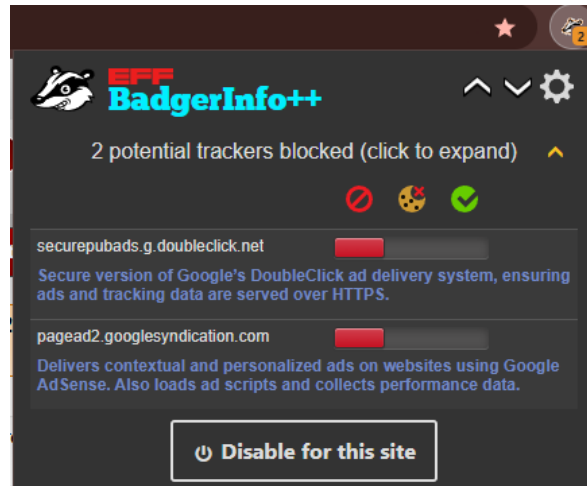


Figure 3.7: The popup window with the individual tracker explanations.

Tracker Explanations

To help users understand what is really going on behind the scenes while they browse, there exists a large internal list of tracker URLs along with short, clear explanations. Each tracker comes with a one-sentence summary that tells who they are and what they are likely to be doing on the site the individual is visiting, whether that is showing ads, tracking behavior or helping the site run. A focus was made on trackers that were detected in the 50 most visited websites in Denmark for 2024, by *Danske Medier* (Official Industry Source in Denmark).

Given the vast and ever-changing landscape of online trackers, it would be neither practical nor effective to include every known tracker in the tool. Instead, a prioritization was made on those that are most prevalent and frequently encountered across the web. This selection strategy ensures that the tool remains efficient and user-friendly while still addressing the majority of tracking activity users are likely to face.

The data was collected and cross-referenced from several trustworthy sources, including WhoTracks.me, DuckDuckGo's Tracker Radar and open-source lists from GitHub and Reddit communities. The team looked at what companies own the trackers, what categories they fall under (like ads, analytics, or social media), and how persistent or invasive they tend to be. Not all of the data made it into the final version; some information was too

technical or contradictory, so the data that was the most useful and easy to explain was utilized.

3.2.6 Internal Page

The second key part of the *BadgerInfo++* extension is the "Internal Page". A web page that can be accessed via the popup and contains detailed information and visualizations about trackers. Uniquely, this page can only be accessed on the highest level of detail display on the popup (one level about the individual URL explanations). At the highest level of detail an underlined "Learn More" text will appear near the bottom of the popup, and when the user clicks on this text, it will open and redirect them to the *Internal Page*. Seen below is what the "Learn More" button looks like.

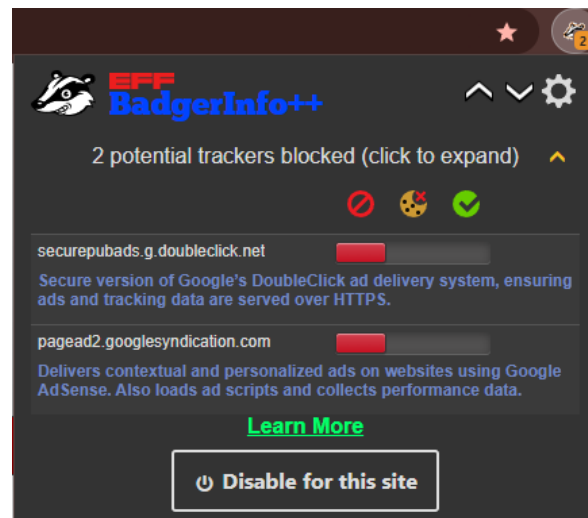


Figure 3.8: The popup window at maximum detail, with the "Learn More" button near the bottom.

The internal page does not rely on a conventional web address, but directly opens an HTML page stored within the extension's files. This page too has multiple "variants" and a way for the user to change the level of detail. The two key parts of this page are the **textual explanations**, and **visual explanations**. By default, **only** the textual explanations are shown, then **only** the visual explanations, and then **both** are shown. Seen below is the internal page in its default state upon being opened.

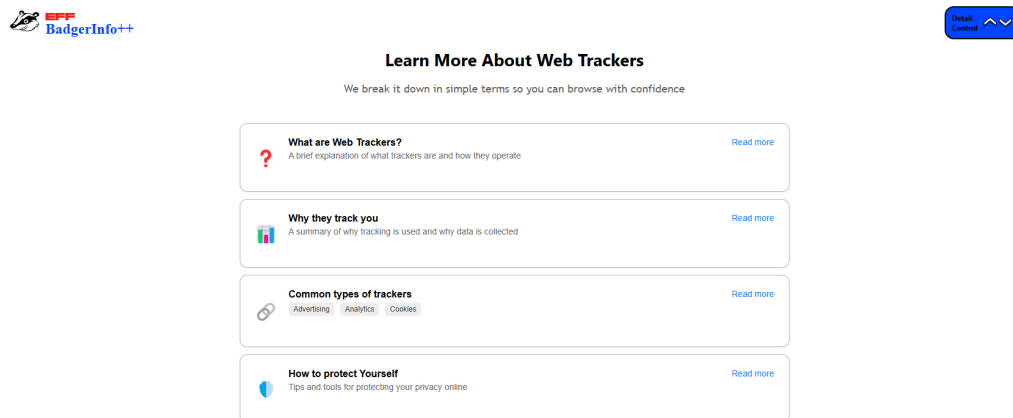


Figure 3.9: The internal webpage, with the default textual explanations only.

Textual Explanations

The internal page contains four sections of textual explanations designed to guide users through the complex subject of web tracking, presented incrementally via "Read More" link. This approach intentionally prevents information overload by allowing users to engage with each section at their own pace. When a user clicks "Read More" the section unfolds to reveal an in-depth explanation of a specific aspect of tracking building a layered understanding of the subject.

Initially, the content establishes a foundational understanding of what trackers are by describing them as snippets of code; such as scripts, images, iframes, or beacons, embedded within webpages. This section explains that these trackers collect various pieces of information, including pages visited, interaction details, device specifications and even location data. Its primary purpose is to clarify the underlying mechanics of web tracking, emphasizing that such data collection is used to build detailed behavioral profiles for targeted advertising, analytics, and user segmentation.

Following this, the guide transitions into an examination of the reasons behind the tracking. It outlines the commercial imperatives, like behavioral advertising and data brokerage, that usually lead widespread tracking practices. By explaining the motivations for both advertisers and website owners, the text provides points for why such tracking exists despite user concerns over privacy. This part of the explanation draws on established research and industry insights.

The third section enters into the technical architecture behind trackers. Explore common methods such as the use of tracking pixels, JavaScript-based tracking, and iframes, as well as more advanced techniques such as cookie and local storage use, fingerprinting, and even CNAME cloaking. Here, each technical mechanism is broken down into simple, digestible explanations that mentions not only how these tools function, but also the potential risks associated with their use. The text ensures that users who are not experts can still grasp the significant points without being mired in overly complex details.

Finally, the last section of the explanation offers practical advice on mitigating tracking. This part discusses how users can protect themselves by adopting privacy-focused browsers, using tracker-blocking extensions, and implementing DNS-based filters. Although it outlines measures such as clearing cookies or disabling JavaScript, it also acknowledges that complete prevention of tracking is challenging. The focus here is on empowering users with actionable steps and tools to reduce unwanted tracking while maintaining website functionality.

Together, these four interlinked sections create an approachable narrative that demystifies web tracking. By structuring the content in layers and allowing progressive disclosure, the design respects the user's learning pace and cognitive load. It not only brings some of the technical and practical aspects of tracking but also aims to embrace a sense of empowerment by informing users.

Internal Tracker Database

Before the visualizations are discussed, it is important to point out a key part of the internals of this page. That being a large **dictionary** of tracker URLs and related information about them.

To identify and annotate known third-party trackers embedded within websites, an internal tracker database was developed. This database serves as the central knowledge base that the tool uses to provide explanatory feedback to users.

The structure of this database was designed to balance readability, flexibility, and performance. It was important for the format to be Human-readable, enabling easy maintenance and manual extension, as well as Machine-parseable, allowing integration with JavaScript-based detection logic.

In parallel, the main objective was offering contextual value and meaningful explanations for users, while keeping it modular and scalable. Additional metadata or attributes could be incorporated in the future without requiring major structural changes.

Each entry in the database corresponds to a single tracker domain and is stored as a JSON object. These objects encapsulate both technical identifiers and human-readable explanations, allowing for dual use in backend logic and frontend display. A representative example of one such entry is as follows:

```
const data = [
  {
    "name": "s.amazon-adsystem.com",
    "explanation": "Amazon's ad system domain for displaying and tracking ads
                  across various websites.",
    "company": "Amazon.com, Inc.",
    "purpose": "Advertising and targeted marketing",
    "category": "Advertising",
    "common_sites": "E-commerce sites, media sites with Amazon ads",
    "privacy_notes": "Tracks user behavior for serving personalized ads and
                     tracking ad performance.",
    "more_info": "https://www.amazon.com/gp/help/customer/display.html?nodeId=201809740"
  },
]
```

Each field within an entry was selected to support both the functional objectives of the tool and its commitment to user education:

- **name:** The exact domain name associated with the tracker. This is used directly in URL or domain matching operations.
- **explanation:** A plain-language summary describing the domain's purpose and activities. This content is displayed in the user interface to aid understanding.
- **company:** The legal entity responsible for the tracker. This introduces transparency and helps attribute behavior to corporate actors.
- **purpose:** A brief description of the tracker's primary function, such as advertising, analytics, or personalization.
- **category:** A broader label grouping the tracker into a recognizable class (e.g., "Advertising" or "Social Media"). Useful for categorization and filtering.
- **tracking type:** Describes the technical nature of the tracking activity, such as behavioral tracking, contextual targeting, or retargeting. This field helps users understand how tracking is performed.
- **common sites:** A free-text field describing the typical environments or website types where the tracker is found. This offers context on the prevalence or strategic placement of the tracker.
- **privacy notes:** Supplementary notes about the tracker's impact on user privacy. This might include behavioral profiling, data retention concerns, or re-identification risks.
- **more info:** A URL linking to the tracker's privacy policy or documentation, allowing users to investigate further.

The chosen flat JSON structure offers a number of key advantages. Firstly, it ensures fast lookup times in browser environments where performance is essential. Secondly, its structure supports direct integration with DOM-parsing scripts and frontend rendering, allowing data from each field to be reused in multiple parts of the application. Thirdly, it allows for easy editing and extension, whether manually or via automated update pipelines derived from public blocklists or corporate disclosures.

Each object is self-contained and includes all the necessary information for both back-end logic and frontend communication. Furthermore, the format is sufficiently flexible to support future enhancements. For instance, localization of explanation fields for international audiences could be incorporated, or new fields such as "cookie duration" or "tracking method" could be added with minimal disruption.

Visual Explanations

After increasing the level of detail from the base *textual explanations* on the internal page, the user will instead see two visualizations (of note: if the user were to increase the detail level again, they would see both the visualizations, and the text). To achieve the visuals on the page, the JavaScript library **d3.js** was used. It is a popular library which can support a wide variety of data visualizations, and is doing the heavy lifting behind the scenes. These two visualizations rely on the internal tracker list from earlier, and provide the user with a new perspective on the trackers they have (knowingly or not) interacted with **since they installed the extension**. If the user did not encounter any trackers before seeing the visualizations, a special message will appear at the top of the internal page informing them of such. This message can be seen in the image below.

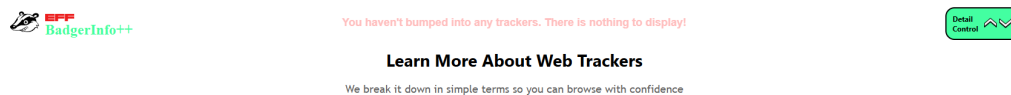


Figure 3.10: Message displayed if the user has not encountered any trackers, and tries to view the internal visualizations.

The first visualization could be most closely compared to a Force-directed graph, although it is more simple in its design. It could broadly be described as a clustered grouping of nodes. Where each "node" represents a single tracker URL and its details. An example of this visualization with a small collection of trackers can be seen below.

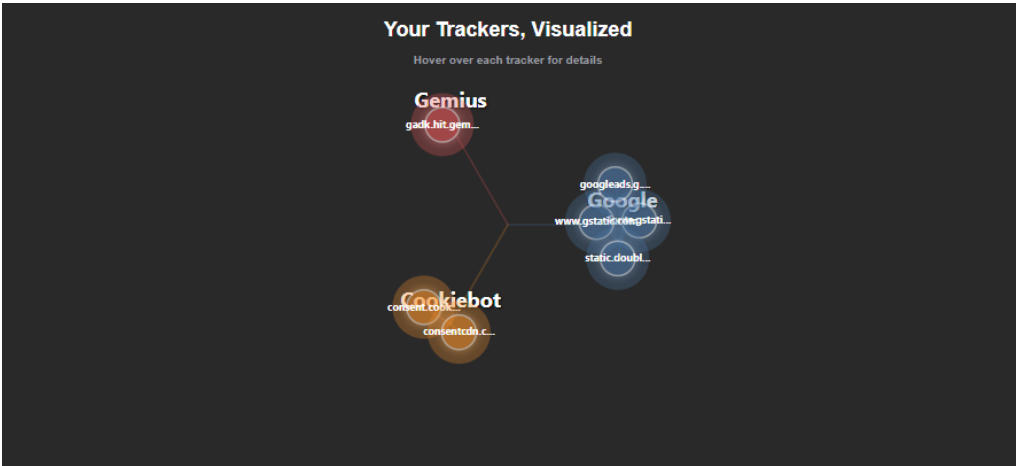


Figure 3.11: The "node graph" visualization with seven trackers.

As can be seen above, the nodes are grouped by whoever owns or operates the tracker, with each group being given a unique color, and a label behind the transparent nodes. The nodes make sure not to overlap with each-other, and have some minor transparency effects and auras for purely aesthetic reasons. One important feature of this visualization and the later visualization is the ability to hover over each element to learn more about it in the form of a popup. In this case, hovering over each node will provide information regarding its; purpose, category, description, and where it is commonly found. This information is sourced from the internal database described earlier. An example of what the popup looks like when hovering over a node can be seen in the image below.

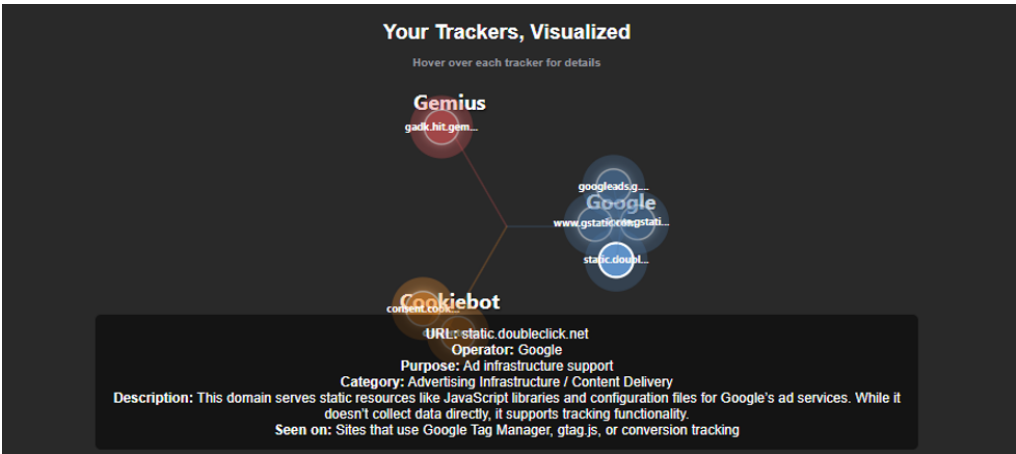


Figure 3.12: The tracker explanation tooltip while hovering over a "tracker node".

With this visual, the user can now have a sense of scale for the amount of trackers they have encountered, along with who owns or operates them.

The second visual is a simple pie chart that groups all the trackers the users has encountered into pie slices based on the category of the tracker. Like the previous visualization, the user can hover over each pie "slice" to learn more information about the group, in this instance, all the tracker URLs that are contained within. While hovering, the specific pie slice will animate and become "raised" from the other slices, which also become more transparent to provide emphasis to the hovered slice. The goal of this visualization was to provide the user with an idea of the variety of trackers they encounter while browsing, and if they wanted more information, they could inspect the specific tracker in the previous visualization. An example setup of the piechart, and its tooltip, can be seen in the two images below.

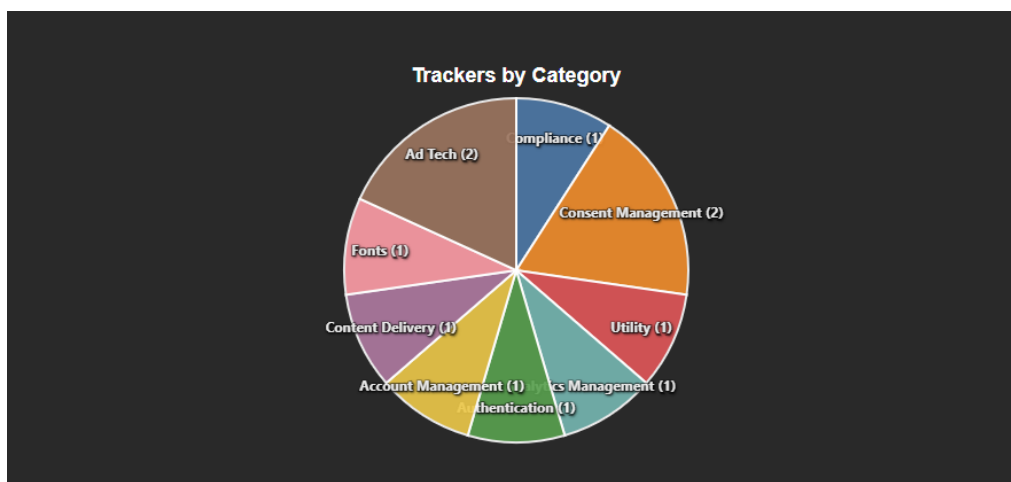


Figure 3.13: The categories pie chart visualization given seven trackers.

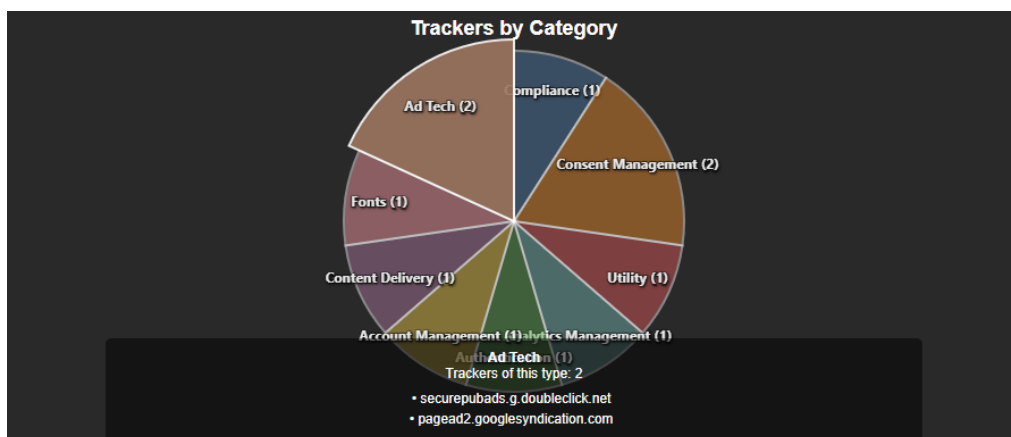


Figure 3.14: The tracker explanation tooltip while hovering over a category "pie slice".

With these two visualizations, the team is satisfied that the user can now have an alternate (and less wordy) perspective of the trackers they have found. Although sometimes not being able to stand on its own without the textual explanations, the user can always choose to hide the visualization by using the **detail** controls. A special note must be made that the team would have liked to include more visualizations to greater enhance the visual understanding of the trackers, but due to time constraints, could not move forward with this vision. It is hoped that the two existing visualizations can stand on their own with the information they provide.

3.2.7 Conclusion

Overall, after reducing the clutter from the original extension, the team is confident that the additional explanations and visualizations introduced in the *BadgerInfo++* extension provide a significantly improved user experience. These enhancements, combined with granular control over how much information is displayed, allow users to tailor their interaction with the extension based on their knowledge and comfort level.

Throughout the development process, the team worked collaboratively to implement an intuitive interface, clear visual indicators, and modular features that enhance user understanding of trackers. The team also addressed several technical issues from the original version, including problems related to element rendering and data flow based on responses from the pilot study. The tracker system was restructured to make it easier to include new entries, ensuring that expanding the database of tracker explanations remains a straightforward process.

While the extension currently covers only a subset of trackers (based on those most commonly seen), the groundwork this project has laid makes future additions efficient and scalable. The team believes that the *BadgerInfo++* extension contributes meaningfully to increasing awareness around online tracking practices, and sees strong potential for continued development and community involvement moving forward.

Chapter 4

Study Design

4.1 Method

The team conducted a semi-structured interview to understand how different types and levels of explanation impact users' understanding of online covert tracking. This approach allowed the researchers to maintain a structured framework of key questions while adapting to participants' responses, enabling the exploration of unexpected themes and insights [8].

Such flexibility is particularly valuable when investigating complex behaviors and perceptions, as it facilitates a deeper understanding of participants' perspectives. The study centered on a modified version of the open-source browser extension **Privacy Badger**, which the team adapted to include several new interface variants that presented tracker-related information in varied formats. These formats ranged from no explanations at all to rich visualizations and detailed text-based descriptions.

The primary objective was to evaluate how these different explanation styles influence users' comprehension, trust, preferences, and awareness when interacting with privacy tools. More specifically, the study aimed to address the research question:

How do different levels (or types) of explanations about covert tracking affect users' understanding and preferences?

To answer this, the research aimed not only to evaluate which interface variants best supported user awareness but also to understand how different user types responded to the same information when presented through distinct design approaches.

4.2 Participants Demographics & Recruitment Process

A total of 38 participants were involved in the study, recruited through sampling methods such as direct outreach, and academic mailing lists. The sample included a mix of

genders, diverse age ranges, with most participants having at least some familiarity with technology use but varied levels of privacy knowledge. This diversity allowed the study to reflect a spectrum of user types—from casual users to those with more technical awareness. Participation was voluntary and conducted under informed consent.

4.3 Study Procedure

A set of open-ended questions was utilized to encourage participants to freely express their opinions while also allowing the interviewer to ask follow-up questions for deeper insights into participants' perceptions. Prior to beginning the interviews, each participant was debriefed on the overall scope of the project, the goals of the study, and the steps involved in the session. Participants were then asked to interact with the locally installed, modified version of the Privacy Badger extension. Due to development constraints, the extension was not published on the Chrome Web Store, and therefore had to be manually installed. This process was guided and explained to each participants at the beginning of the session.

Once installed, the user was instructed to visit several websites of their choice to simulate typical browsing behavior. During this activity, they interacted with all four interface variants embedded in the extension, each offering a different level or type of explanation related to web tracking:

- **Variant A:** No additional information beyond tracker blocking.
- **Variant B:** Inline text explanations under each blocked tracker.
- **Variant C:** A “Learn More” button that revealed expanded explanations and definitions.
- **Variant D:** A visual display (graph and pie chart) showing how trackers are interconnected.

Interview questions were organized into two categories - *during use* and *after use of the extension*. The *during use* questions focused on capturing participants' immediate reactions and understanding while interacting with the interface, whereas the *after use* questions aimed to explore their overall impressions, preferences, and suggestions following the full experience. Participants were guided through the use of each variant, asked to reflect on their understanding and preferences, and then interviewed using a semi-structured format. The interviews were conducted individually and focused on evaluating each participant's perceptions, comprehension, and emotional response.

4.4 Findings & Discussion

4.4.1 Participant Preferences: Identifying the Best Variant

Participants were exposed to four different UI variants of the modified Privacy Badger extension, each designed to convey tracker activity with varying degrees of depth and complexity. Analysis revealed that participants' preferences were diverse, often reflecting their personal comfort levels with visual information, reading complexity, and interest in

privacy issues.

Variant D (Visual analytics) emerged as the most engaging for tech-savvy users, offering interactive and intuitive graphs that made covert tracking more tangible. Users appreciated the visual representation of tracker networks and types, describing it as "fun," "eye-opening," and "empowering." For example, P02 stated, *"The graph was really interesting. I could instantly see which websites were tracking me and how they were connected."* Similarly, P10 noted, *"I liked the visual style because I could immediately understand how trackers were linked across different sites."* However, several users also noted that the visuals were occasionally cluttered or difficult to interpret. P08 mentioned, *"Some of the circles in the graph overlapped too much - it was hard to tell which tracker belonged where."*

Variant C (Learn More deep-dive) appealed strongly to those seeking contextual understanding. This version offered layered explanations, definitions, and links to external resources. It was frequently described as informative and digestible, particularly for non-technical users. P07 reflected, *"The Learn More button helped me understand what a tracker actually is. Without it, I'd just be guessing."* P14 added, *"Having the option to read more, rather than being forced to, made it easier to absorb."*

Variant B (Inline text) was praised for its simplicity and clarity. Participants who preferred straightforward, minimal disruption favored this version, often calling it "just enough" information. P27 shared, *"The little text under each tracker was perfect - I didn't need to click around or read a lot."*

Variant A (No new info) was the least preferred across participants, with most users perceiving it as lacking transparency or depth. P12 stated, *"It blocked the trackers, sure, but I had no idea what was going on behind the scenes. I felt left out."*

Overall, preferences aligned with user type: casual users preferred clarity and brevity (Variant B), curious or non-technical users preferred guided exploration (Variant C), and tech-savvy users leaned toward visual, data-rich content (Variant D).

4.4.2 Awareness of Tracking Post-Use

The study demonstrated a notable increase in participants' awareness of web tracking after using the extension. Many reported a clearer understanding of what trackers are, how they function, and the extent of data collection across the web. Several users noted behavioral changes, such as becoming more cautious about cookies or questioning default website settings.

P30 remarked, *"I didn't realize how many trackers are on just one website. It really opened my eyes."* P13 shared, *"I used to accept all cookies. Not anymore."* P13 also expressed a greater sense of caution: *"Now I'm suspicious of every website I visit. I think twice before I click anything."*

The extension, regardless of variant, served as a catalyst for enhanced user understanding, particularly when explanations were layered or visually supported. P31 emphasized, *"The extension helped me understand not just that I was being tracked, but how it works - like what data*

they're actually collecting."

Beyond increased awareness, several participants also indicated a likelihood of adjusting their online behavior in response to what they learned. Some described a growing habit of rejecting unnecessary cookies or being more cautious about privacy settings. *P11 said, "Now I don't just click 'accept all' anymore, I check every setting before continuing. P25 shared, "I'll definitely start looking into other tools that help protect my data — this was a wake-up call."* These statements reflect an emerging behavioral intention to adopt privacy-conscious habits moving forward, suggesting that the tool's influence may extend beyond the immediate user session.

4.4.3 Perceptions of Information Level

Participants generally found the level of information provided to be appropriate, especially when it was adjustable or layered. Variant C's expandable definitions and Variant D's visual graphs allowed users to self-select the depth of detail they engaged with.

However, some noted that terminology such as "domain," "CDN," or "persistent" was overly technical. *P11 commented, "Some of the words confused me - like I didn't know what a CDN was. It needs a glossary or something simpler."* Visual complexity in Variant D was occasionally cited as overwhelming or visually confusing, particularly where data nodes overlapped. *P08 explained, "It was hard to read the graph - everything was just too close together."*

Participants valued clarity, readability, and flexibility. There was a clear preference for UI designs that allowed users to explore deeper content without forcing it upfront. *P16 shared, "I liked having the basic info there, but I also liked that I could dig deeper when I wanted to. That balance really worked for me."*

4.4.4 Participant Suggestions for Improvement

Participants provided a wide range of suggestions, categorized below:

Functionality:

- Include tutorials or onboarding flows: *P15 suggested, "A short intro video would help new users figure out how to use it."*
- Confirm when trackers are successfully blocked: *P28 said, "I wanted a message that told me the tracker was actually blocked - not just listed."*
- Offer more context about trackers: *P07 said, "Who even owns these trackers? I wanted to know if they were safe or shady."*
- Incorporate a trustworthiness indicator: *P13 humorously suggested, "Add a leaderboard of creepiest trackers - that'd be fun and informative."*

Design and UI:

- Simplify visual presentations: *P08 commented, "The graphs were interesting but messy. I couldn't really read all of it."*
- Improve readability: *P31 noted, "Some fonts were small and hard to read. Bigger text would be better."*
- Better formatting for Learn More: *P11 recommended, "Make it expandable or collapsible. Too much text on one page is a bit much."*
- Add dark mode: *Mentioned by P17, who said, "The light theme was too bright. Dark mode would be easier on the eyes."*

Accessibility and Clarity:

- Use simpler language: *P31 emphasized, "Avoid jargon. Not everyone is into tech."*
- Provide audio or video aids: *P19 said, "An explainer video would be really useful for people who don't want to read."*
- Add multi-language support: *P25 mentioned, "This would be great for non-English speakers too."*

Participants appreciated having access to tracker data but emphasized that the tool should educate without overwhelming. These suggestions underscore the importance of designing for inclusiveness and usability across different user types.

4.5 Discussions

The findings of this study provide significant insight into how different explanation styles within a tracker-blocking extension influence user understanding, preferences, and behaviors. Each version (A-D) provided unique benefits, and participants' replies showed that explanation style has a direct impact on both engagement and comprehension.

Variants C and D, which provided layered text-based information and visual analytics respectively, were most effective in increasing user awareness. These formats allowed users to engage at their own pace and depth, suggesting that flexibility in information presentation enhances learning. Variant C appealed to non-technical users who preferred guidance and clarity, while Variant D engaged more tech-savvy participants who found the visualizations intuitive and empowering. These results suggest that effective privacy tools must accommodate different user preferences by offering multi-layered or customizable experiences.

Moreover, the reported behavioral intentions post-use, such as greater caution in cookie acceptance and interest in exploring additional privacy tools, indicate that user education within interface design can have a lasting impact on digital behavior. This aligns with research in HCI and persuasive design, which shows that clear, contextualized information delivery can foster more privacy-conscious habits.

Interestingly, the study also uncovered the role of trust and emotional response. Some participants expressed uncertainty about whether trackers were truly being blocked, emphasizing the importance of trust signals in user interface design. Others experienced discomfort or surprise upon discovering the extent of tracking, highlighting that emotional response can be a critical lever in privacy education.

Although these findings align with existing literature that emphasizes transparency and user agency [15], this study contributes by showing how the format of transparency, not just its presence, can shape outcomes. Users are not uniformly responsive to the same type of information; thus, privacy tools must be able to adapt to varying cognitive styles and technical backgrounds.

However, there are limitations to consider. The study was based on a limited sample size and included participants from relatively similar educational backgrounds. Furthermore, the subjective nature of qualitative interviews can introduce interpretation bias. These limitations point to the need for larger and more diverse future studies that explore similar interventions across different demographics.

In general, the study illustrates that, when designed thoughtfully, browser extensions can not only enhance technical privacy but also empower users through increased understanding. The feedback gathered provides a strong foundation for iterating on educational interface design and offers a compelling direction for future development of privacy-focused tools.

4.6 Future Work

The project's extension, and performed study, has a wide room for expansion and addition for future work.

- While this study identified self-reported changes in awareness and behavior, a future longitudinal study could track actual changes in browsing habits over time. This might include examining cookie rejection rates, usage frequency of privacy tools, or shifts in interaction with ad content.
- One significant hurdle encountered by participants during the study was the manual installation process, as the modified extension was not publicly available on the Chrome Web Store. Developing an official standalone version of the extension would substantially improve accessibility, streamline installation, and facilitate broader participation recruitment in future studies.
- Extension development was focused on only supporting the *Chrome* web browser. Future efforts could be made to add compatibility with other web browsers like Firefox, Brave, Opera, Safari, etc.
- The individual explanations provided to each tracker URL in the extension was quite limited, as though the more than 100 explanations covered the most frequent trackers, there exist thousands on the web which awkwardly did not have an explanation

for the user to see. Though a potentially Sisyphean task, adding more individual tracker URL explanations could greatly improve the quantity of information the user has access to.

- Incorporating short explanation videos or onboarding tutorials, could help users better understand tracking and privacy concepts, especially those new to digital safety tools.
- Several participants commented on the cluttered or overlapping visuals in Variant D. A focused study on improving data visualization in privacy tools, including color contrast, animation, and interactivity, would be valuable.
- Future iterations could explore adaptive interfaces that tailor the depth and format of tracking information based on a user's technical proficiency or interaction history.
- Given user interest in knowing whether trackers are "safe" or "sketchy," future versions of the extension could incorporate visual trust indicators, credibility ratings, or labels (e.g., known ad networks vs. benign analytics).

By pursuing these lines of inquiry, researchers and developers can further enhance the usability, impact, and inclusivity of privacy-centered tools like the modified extension explored in this study.

Chapter 5

Conclusion

5.1 Problem Formulation and Solution

This project addressed a growing concern in the digital landscape: users' limited understanding of how online tracking works and how it fuels targeted advertising. Despite the availability of privacy-enhancing tools like tracker blockers, many users still feel disempowered or unaware of how their data is collected and used.

The central research question is: *How do different levels (or types) of explanations about covert tracking affect users' understanding and preferences?* This question guided the design and evaluation of a modified browser extension called **BadgerInfo++**.

Built on top of **Privacy Badger**, the extension introduced layered explanations in both textual and visual formats to improve user awareness and interaction. Through a semi-structured interview study with **38** participants, the evaluation explored how these different explanation types impacted users' understanding, trust, and behavioral intentions.

Findings show that while there is no one-size-fits-all explanation style, users benefit from being able to choose the level and format of information they engage with. Textual summaries were preferred by non-technical users for their clarity, while visual analytics resonated more with tech-savvy participants. Overall, the combination of layered explanation, interface flexibility, and interactive elements proved effective in helping users better comprehend the mechanics of online tracking.

5.2 Final Thoughts

BadgerInfo++ demonstrates that educational enhancements within privacy tools can meaningfully improve user understanding, trust, and decision-making. Rather than simply blocking trackers in the background, the extension engages users with contextual, digestible information that promotes reflection and encourages more privacy-conscious behavior.

While the study had limitations such as a small, relatively homogenous participant pool and technical constraints in deployment, it offers valuable insights for the future of privacy tool design. By embracing personalization, multimodal explanations, and layered transparency, privacy tools can move beyond protection alone and play a vital role in user education.

Ultimately, this project underscores the importance of designing for both technical function and user comprehension. As digital privacy continues to evolve, tools like **BadgerInfo++** represent a promising step toward improving transparency and supporting user agency in digital environments.

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Appendix A

Project Code

A.1 Code

The entire codebase for this project can be found on our github repository.

A.2 Installation

To install the extension, follow these instructions:

1. On the github page, click on the green "Code" button, and download the entire repository as a zip file.
2. Unzip the downloaded folder and extract all content.
3. Open the **Chrome** browser.
4. Click on the 3 vertical dots at the top right corner to open the menu.
5. Click on *Extensions* → *Manage Extensions*.
6. At the top right corner, enable "*Developer Mode*".
7. At the top left corner, select "Load unpacked".
8. Navigate to the "**src**" folder under the "CSIT8-DWTA-main" and hit the "select folder" button.
9. The extension should now be automatically installed under "All extensions" as *BadgerInfo++*.

The first time installation page will appear after a few seconds.

Appendix B

Google Form for Participant Solicitation

The following **Google Form** was created and shared via email to Aalborg University students in order to solicit participants:

B.1 Google Form Content

Help Us Improve Online Privacy!

We're conducting a short study in connection with our project for the 8th semester (Dealing with Targeted Advertisements) to understand how people perceive online tracking and targeted advertisements. This form involves answering some few questions in relation to the study after which you will be contacted for the study session. The study session involves installing a browser extension, browsing a few websites, and answering some questions. It'll only take some few minutes. Your input and experience will be crucial in making privacy tools more user-friendly and educational!

No technical knowledge required!

If you're interested, please fill out the form below.

All data collected will be anonymous and used for academic purposes only.

1. Gender*

- Male
- Female
- Other

2. Age*

- 18 - 24
- 25 - 34
- 35 - 44
- 45+

3. What is your occupation or field of study?*

4. Are you currently using any privacy-related tools (like ad blockers or browser extensions)?*

- Yes
- No

5. How often do you think about your privacy when browsing the web?

- Never
- Rarely
- Occasionally
- Sometimes
- Frequently
- Often
- Always

6. How concerned are you about websites tracking your online behavior and thereby getting information about you?*

- Very unconcerned
- Unconcerned
- Slightly unconcerned
- Neutral
- Slightly concerned
- Concerned
- Very concerned

7. How familiar are you with the concept of online trackers that collect your browsing data?

- Very unfamiliar
- Unfamiliar
- Somewhat unfamiliar
- Neutral
- Somewhat familiar
- Familiar
- Very familiar

8. What do you think about ads that are shown to you based on the websites you visit?*

- I find them useful
- I don't mind them
- I find them annoying
- I don't like them at all

9. Would you be willing to participate in a session that involves installing a browser extension and sharing your experience? (It'll only take some few minutes)*

- Yes
- No

Appendix C

Interview Structure

C.1 Consent for Interview participation

Voluntary Participation

Participation in this study is completely voluntary. You may choose to withdraw from the study at any time, without any consequences. If you decide to withdraw, your data will not be included in the final analysis.

Confidentiality

Your responses will be kept confidential and will be used solely for the purposes of this study. All responses will be anonymized, and no personal identifiers will be included in any reports or publications.

Audio Recording

Please note that audio will be recorded during the interview/feedback session that follows this survey. The recording will be used only for analysis purposes to help us better understand your experience and feedback. The audio will be kept confidential, and no identifiable information will be linked to the recordings in any reports or publications. You may ask to stop the audio recording at any point during the session.

Data Use

By agreeing to participate, you give us permission to use your responses (and the audio recording, if applicable) to evaluate the effectiveness of the browser extension. Your information will not be shared with third parties, and your privacy will be respected throughout the study.

C.2 Interview Questions

C.2.1 During Use of the Extension

- Q1.** Does the extension notification or messages stand out to you?
- Q2.** Do you understand what the tracker information is telling you?
- Q3.** Is there any information about the trackers that you would have liked to see but wasn't included?

C.2.2 After Use of the Extension

- Q4.** Which variant did you find most helpful or informative? Why?
- Q5.** Did you find it easy to understand how the information about trackers was displayed?
- Q6.** What do you think about the amount of information provided about the trackers?
- Q7.** Did the extension help you understand what trackers are and how they work?
- Q8.** Do you think the information provided by using the extension has helped you increase your awareness of the trackers and ads on the websites you visit?
- Q9.** Would you be more likely to continue using a tool like this?
- Q10.** What do you think we can improve about the extension?